

Student Guide for Sync Session

Week 8: Data Visualisation in Power BI

This guide will help you navigate the session effectively. It includes key insights and strategies to enhance your learning experience. Stay engaged and make the most of this session!

Session Overview

Session Title	Data Visualisation in Power BI	
Session Duration	3 hours	
Session Type	Lectures: Conceptual understanding of Power BI Visuals Case Studies: Real-world applications of visualisation in business analytics	
Scope	This session covers fundamental visualisation techniques in Power BI and their role in data analysis, including: - Importance of data visualisation - Key Power BI visuals: column charts, bar charts, card visuals, pie charts, maps, gauge charts - Custom visuals and formatting in Power BI - Case study applications in business analytics	
Learning Objectives		
	Objective	Core Capability
	Understand the role of visualisation in data analysis	Foundational knowledge in analytics
	Apply key Power BI visuals for effective reporting	Hands-on skills in Power BI
	Customise and format Power BI visuals	Enhancing data presentation skills
	Implement case study applications for business insights	Practical application of visualisation
Software/Tools	Power BI	

Pro Tips for Success

- Think Visually:** Choose the right chart type for your data.
- Keep it Clean:** Avoid clutter and focus on key insights.
- Practice Hands-On:** Experiment with different Power BI visuals to understand their impact.
- Engage in Discussions:** Share your interpretations and insights with peers.

Session Details

Topic/Concept	A Glimpse	Insights/Actionable
Introduction to Data Visualisation	Transforming raw data into insights	Understand how visuals enhance decision-making

Importance of Power BI Visualisation	Breaking down complex data into simple visuals	Identify key benefits such as spotting trends and patterns
Column and Bar Charts	Comparing data across categories or time periods	Learn to visualise differences and analyse trends
Card Visuals & Multi-Row Cards	Tracking key metrics effectively	Display important business metrics in dashboards
Pie and Doughnut Charts	Showing part-to-whole relationships	Apply these visuals for sales distribution analysis
Maps in Power BI	Representing geographic data effectively	Learn how to visualise data by region
Gauge Charts	Tracking progress against targets	Use for performance monitoring in businesses
Custom Visuals in Power BI	Expanding visualisation capabilities	Import and use custom visuals from the Power BI store
Formatting Visuals	Enhancing clarity and design	Customise colors, labels, legends, and other elements
Case Study: Sales Performance Analysis	Applying visualisation techniques to analyse business metrics	Develop insights using Power BI dashboards
Summary and Q&A	Recap of key takeaways and interactive discussion	Reinforce learning through discussion

Post-Session Activities

Reflection Challenge	Think about a dataset you have worked with. How could visualisation improve its analysis? Write a short summary.
Explore More	Read: Experiment with Power BI visuals on different datasets. Watch: Research best practices for dashboard design. Play around: Learn about Power BI's AI-powered insights.
Get Inspired	Read case studies on data-driven decision-making using Power BI. Watch tutorials on advanced visualisation techniques.
The Journey Ahead	Apply Power BI visualisation in your projects. Continue learning about interactive dashboards and real-time data analytics.